

TidyTuesday Week 39

Pranay Gundam

Friday 25th September, 2026

Contents

1	Weekly Summary	2
2	Date: 2026-09-21	4
2.1	Raw Regression	4
2.2	Detrended Regression	6
3	Date: 2026-09-22	8
3.1	Raw Regression	8
3.2	Detrended Regression	10
4	Date: 2026-09-23	12
4.1	Raw Regression	12
4.2	Detrended Regression	14
5	Date: 2026-09-24	16
5.1	Raw Regression	16
5.2	Detrended Regression	18
6	Date: 2026-09-25	20
6.1	Raw Regression	20
6.2	Detrended Regression	22

1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the weekly parade of numbers that have no business being in the same room together, let alone the same scatterplot. This week our little algorithm reached into the FRED grab bag and pulled out New England electricity prices, house listings in Flathead County Montana, wholesale payrolls, some discontinued bank asset ratio, and — my personal favorite — the number of tax exemptions claimed by people under 65 in New Mexico. As always, none of these things asked to be compared to each other. We did it anyway.

Monday kicked us off with the textbook trend mirage. New England commercial electricity prices versus a medical services price index posted a jaw-dropping R-squared of 0.93 in the raw regression, which, if you didn't know better, would have you drafting a Nobel acceptance speech. Then we detrended it and the R-squared cratered to a heroic 0.02 with a p-value of 0.51, which in technical terms means "absolutely nothing is happening here." Two things went up over two decades, as prices tend to do, and the raw regression got very excited about it. Case closed, mirage confirmed, everyone go home.

But then the week got genuinely weird, in that some of these actually held up. Tuesday's pairing of Flathead County active home listings with U.S. imports of maintenance and repair services survived detrending — it actually got a bit stronger, jumping to a still-modest R-squared around 0.19 with a p-value in the "huh, that's real" range. Wednesday was the standout: wholesale trade payrolls versus the Dover, Delaware house price index kept a chunky R-squared of 0.73 even after stripping out the shared trend, which is the kind of result that makes me squint suspiciously at my own code. Thursday's Southwest central-bank GDP versus that discontinued Pacific bank ratio also clung to significance detrended (R-squared crept up to 0.28, and negative, no less), and Friday's education-and-communication CPI versus New Mexico tax exemptions doubled its R-squared to a frankly show-offy 0.51 once detrended. Four out of five survivors is an unusually strong week for a project whose entire premise is finding nonsense.

Now, before anyone starts building a hedge fund around Delaware home prices and wholesale payroll indexes, let me remind you with all the deadpan sincerity I can muster: these are randomly paired series. I did not have a theory. There is no theory. Nobody is claiming that New Mexico tax filers are secretly driving the price of your kid's textbooks, and nobody sane would. A relationship "holding up under detrending" makes it more interesting than the electricity-price fiasco, not actually meaningful — it just means the coincidence is a little more committed to the bit. Spurious correlation is the house specialty here, served fresh daily, and this week's suspiciously robust survivors are best enjoyed the way you'd enjoy a horoscope that happened to be

technically accurate. See you next week for more statistically confident gibberish.

2 Date: 2026-09-21

Series ID: PCU221122221122421

This series is titled Producer Price Index by Industry: Electric Power Distribution: Commercial Electric Power for New England Census Division and has a frequency of Monthly. The units are Index Dec 1990=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1971-01-01 and the observation end date is 2026-08-01. The popularity of this series is 1.

Series ID: SYSIILPIBLEND

This series is titled Medical Services Expenditures by Disease: Symptoms; Signs; and Ill-Defined Conditions Price Index, Blended Account Basis and has a frequency of Annual. The units are Index 2017=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2000-01-01 and the observation end date is 2021-01-01. The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_SYSIILPIBLEND	R-squared:	0.933
Model:	OLS	Adj. R-squared:	0.929
Method:	Least Squares	F-statistic:	277.7
Date:	Mon, 21 Sep 2026	Prob (F-statistic):	3.41e-13
Time:	17:16:08	Log-Likelihood:	-68.300
No. Observations:	22	AIC:	140.6
Df Residuals:	20	BIC:	142.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-6.0876	5.281	-1.153	0.263	-17.103	4.928
value_fred_PCU221122221122421	0.5458	0.033	16.664	0.000	0.477	0.614

Omnibus:	6.881	Durbin-Watson:	1.268
Prob(Omnibus):	0.032	Jarque-Bera (JB):	5.911
Skew:	0.534	Prob(JB):	0.0521
Kurtosis:	5.304	Cond. No.	706.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

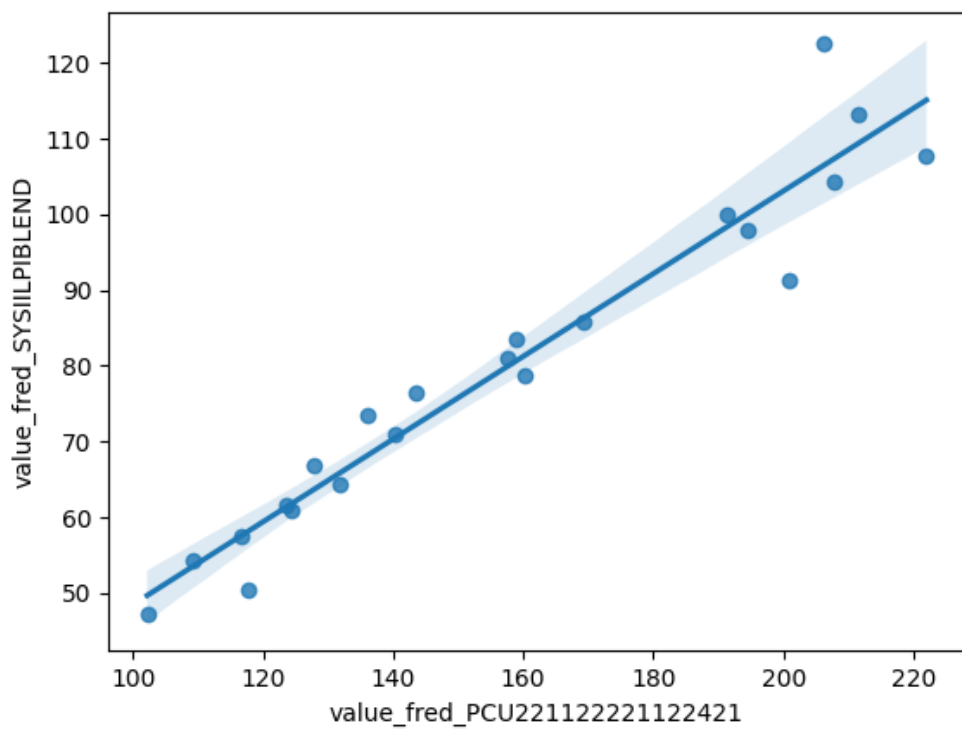


Figure 1: Raw Regression Plot for 2026-09-21

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_SYSIILPIBLEND_detrended	R-squared:	0.022
Model:	OLS	Adj. R-squared:	-0.027
Method:	Least Squares	F-statistic:	0.4428
Date:	Mon, 21 Sep 2026	Prob (F-statistic):	0.513
Time:	17:16:08	Log-Likelihood:	-54.441
No. Observations:	22	AIC:	112.9
Df Residuals:	20	BIC:	115.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	4.119e-14	0.643	6.41e-14	1.000	-1.340	1.340
value_fred_PCU221122221122421_detrended	0.0480	0.072	0.665	0.513	-0.103	0.199

Omnibus:	12.738	Durbin-Watson:	0.457
Prob(Omnibus):	0.002	Jarque-Bera (JB):	12.678
Skew:	1.200	Prob(JB):	0.00177
Kurtosis:	5.841	Cond. No.	8.90

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

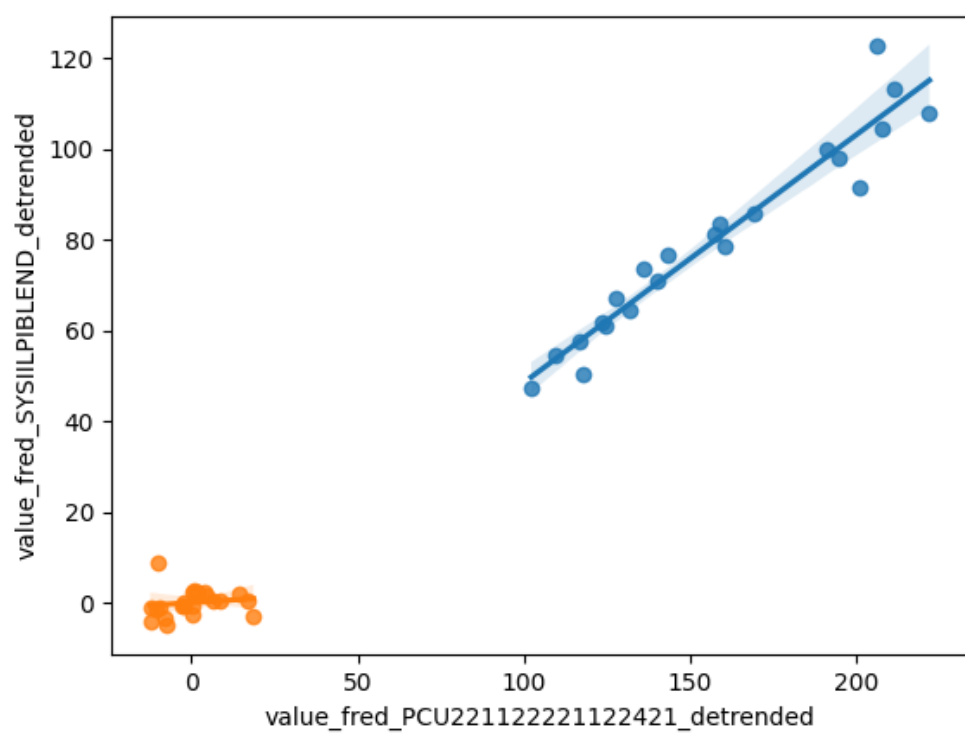


Figure 2: Detrended Regression Plot for 2026-09-21

3 Date: 2026-09-22

Series ID: ACTLISCOU30029

This series is titled Housing Inventory: Active Listing Count in Flathead County, MT and has a frequency of Monthly. The units are Level and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2016-07-01 and the observation end date is 2026-08-01. The popularity of this series is 2.

Series ID: ITMMARM133S

This series is titled U.S. Imports of Services: Maintenance and Repair Services, not included elsewhere and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Seasonally Adjusted. The observation start date is 1999-01-01 and the observation end date is 2026-07-01. The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_ITMMARM133S	R-squared:	0.123
Model:	OLS	Adj. R-squared:	0.116
Method:	Least Squares	F-statistic:	16.76
Date:	Tue, 22 Sep 2026	Prob (F-statistic):	7.76e-05
Time:	15:49:46	Log-Likelihood:	-747.35
No. Observations:	121	AIC:	1499.
Df Residuals:	119	BIC:	1504.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	452.9830	36.286	12.484	0.000	381.134	524.832
value_fred_ACTLISCOU30029	0.1837	0.045	4.094	0.000	0.095	0.273

Omnibus:	12.594	Durbin-Watson:	0.214
Prob(Omnibus):	0.002	Jarque-Bera (JB):	14.022
Skew:	0.833	Prob(JB):	0.000902
Kurtosis:	3.056	Cond. No.	2.75e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.75e+03. This might indicate that there are strong multicollinearity or other numerical problems.

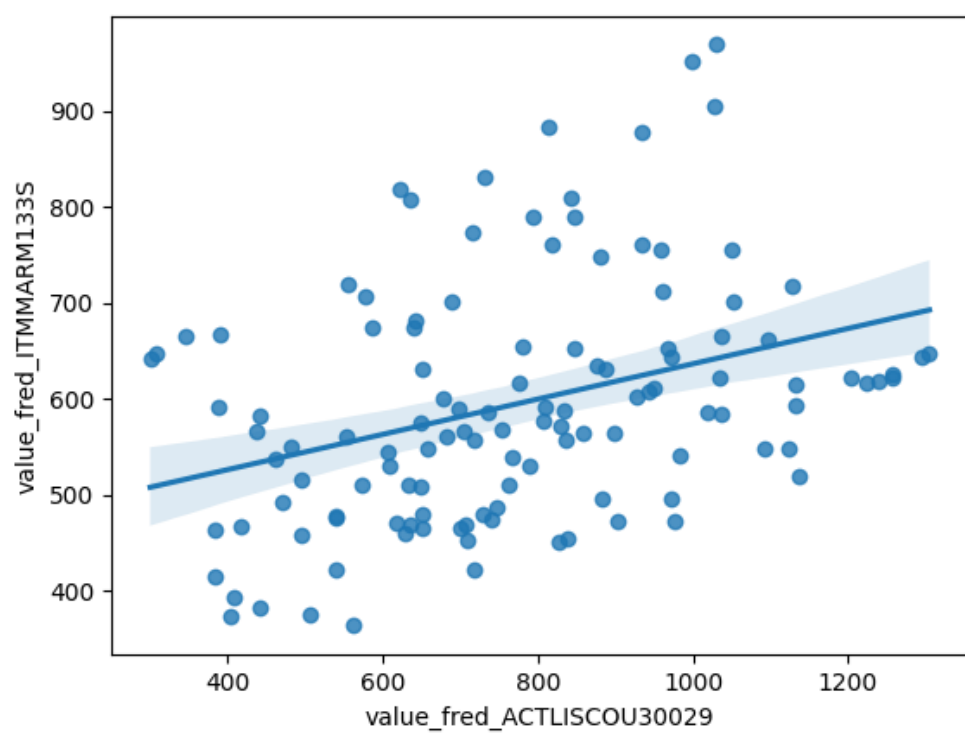


Figure 3: Raw Regression Plot for 2026-09-22

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_ITMMARM133S_detrended	R-squared:	0.195
Model:	OLS	Adj. R-squared:	0.188
Method:	Least Squares	F-statistic:	28.77
Date:	Tue, 22 Sep 2026	Prob (F-statistic):	4.07e-07
Time:	15:49:46	Log-Likelihood:	-740.70
No. Observations:	121	AIC:	1485.
Df Residuals:	119	BIC:	1491.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-3.037e-11	10.104	-3.01e-12	1.000	-20.008	20.008
value_fred_ACTLISCOU30029_detrended	0.2437	0.045	5.364	0.000	0.154	0.334

Omnibus:	4.932	Durbin-Watson:	0.254
Prob(Omnibus):	0.085	Jarque-Bera (JB):	5.006
Skew:	0.473	Prob(JB):	0.0819
Kurtosis:	2.689	Cond. No.	222.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

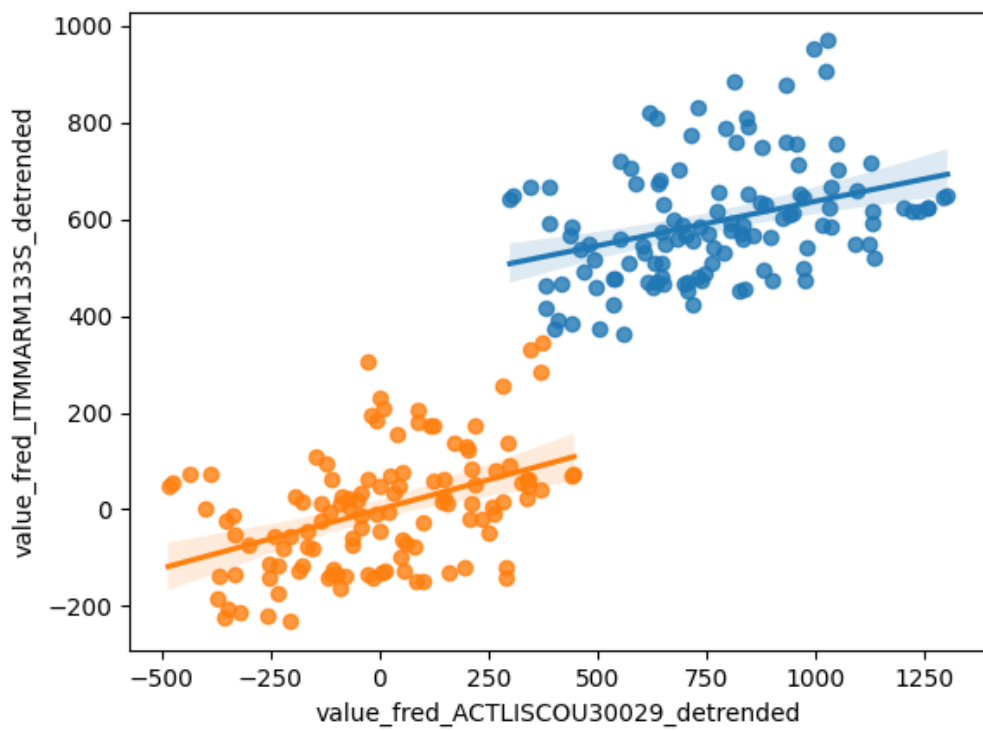


Figure 4: Detrended Regression Plot for 2026-09-22

4 Date: 2026-09-23

Series ID: CEU4142000017

This series is titled Indexes of Aggregate Weekly Payrolls of All Employees, Wholesale Trade and has a frequency of Monthly. The units are Index 2007=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2006-03-01 and the observation end date is 2026-08-01. The popularity of this series is 1.

Series ID: ATNHPIUS20100Q

This series is titled All-Transactions House Price Index for Dover, DE (MSA) and has a frequency of Quarterly. The units are Index 1995:Q1=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1987-07-01 and the observation end date is 2026-04-01. The popularity of this series is 2.

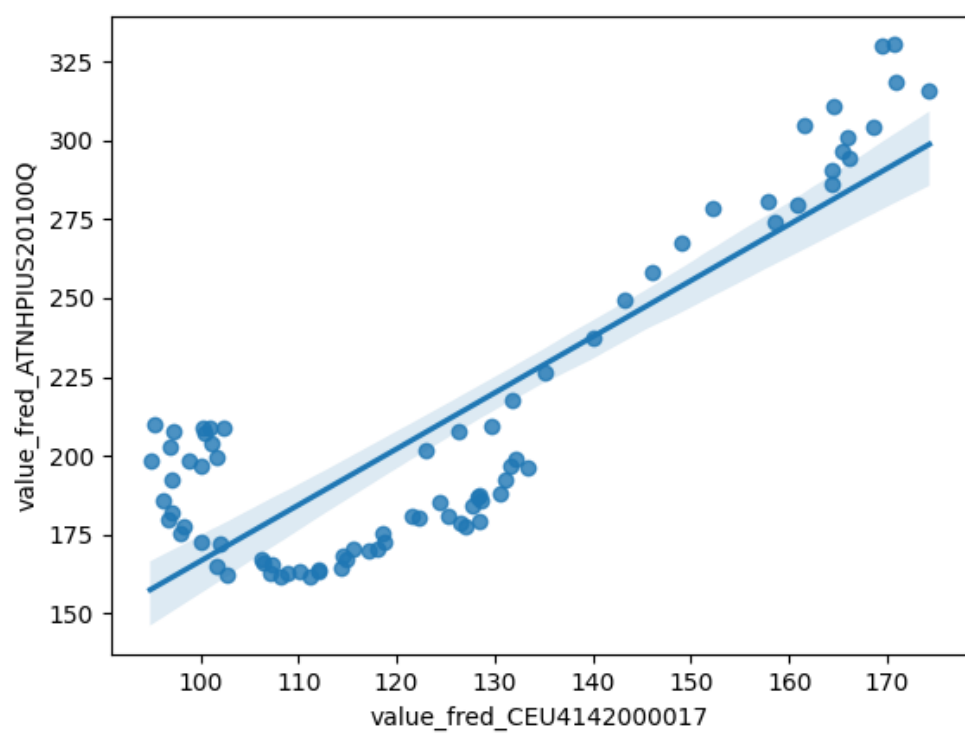
4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_ATNHPIUS20100Q	R-squared:	0.731			
Model:	OLS	Adj. R-squared:	0.728			
Method:	Least Squares	F-statistic:	214.8			
Date:	Wed, 23 Sep 2026	Prob (F-statistic):	3.06e-24			
Time:	15:39:04	Log-Likelihood:	-377.92			
No. Observations:	81	AIC:	759.8			
Df Residuals:	79	BIC:	764.6			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	-11.5081	15.435	-0.746	0.458	-42.230	19.214
value_fred_CEU4142000017	1.7805	0.121	14.656	0.000	1.539	2.022
Omnibus:	32.589	Durbin-Watson:	0.083			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	6.325			
Skew:	0.272	Prob(JB):	0.0423			
Kurtosis:	1.744	Cond. No.	678.			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_ATNHPIUS20100Q_detrended	R-squared:	0.727
Model:	OLS	Adj. R-squared:	0.724
Method:	Least Squares	F-statistic:	210.7
Date:	Wed, 23 Sep 2026	Prob (F-statistic):	5.32e-24
Time:	15:39:04	Log-Likelihood:	-349.22
No. Observations:	81	AIC:	702.4
Df Residuals:	79	BIC:	707.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-7.785e-12	2.029	-3.84e-12	1.000	-4.039	4.039
value_fred_CEU4142000017_detrended	4.4163	0.304	14.517	0.000	3.811	5.022

Omnibus:	7.128	Durbin-Watson:	0.420
Prob(Omnibus):	0.028	Jarque-Bera (JB):	3.489
Skew:	0.261	Prob(JB):	0.175
Kurtosis:	2.128	Cond. No.	6.67

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

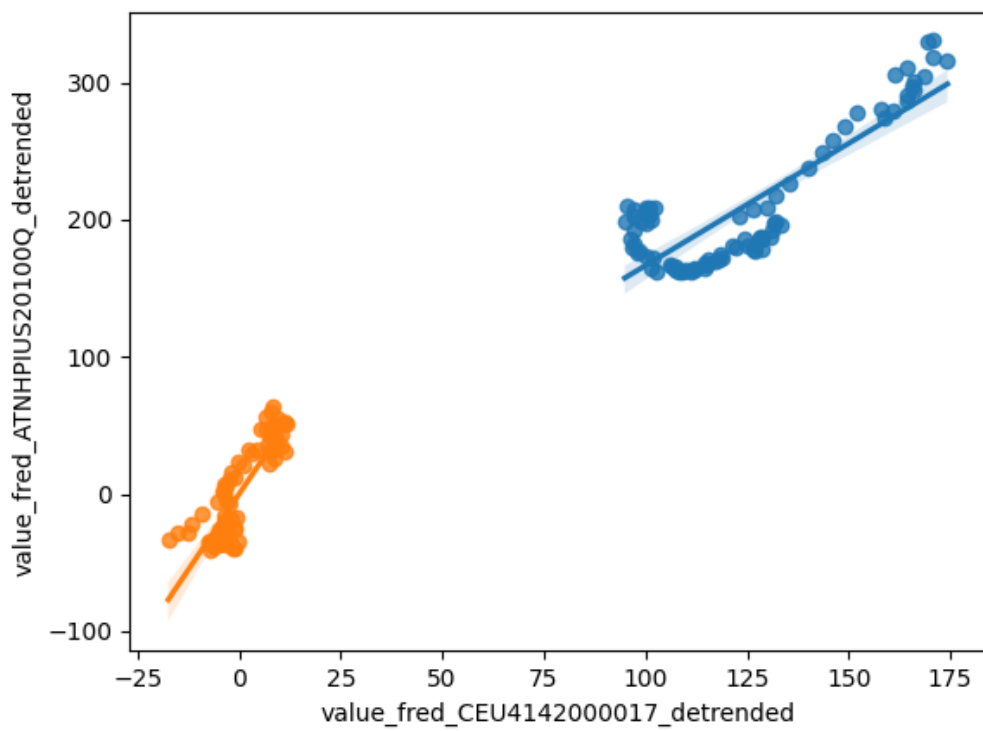


Figure 6: Detrended Regression Plot for 2026-09-23

5 Date: 2026-09-24

Series ID: SWSTFRBCIRGSP

This series is titled Real Gross Domestic Product: Monetary Authorities-Central Bank, Credit Intermediation, and Related Services (521-522) in the Southwest BEA Region and has a frequency of Annual. The units are Millions of Chained 2017 Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1997-01-01 and the observation end date is 2024-01-01. The popularity of this series is 1.

Series ID: LLRNPT39

This series is titled Assets at Banks whose ALLL exceeds their Nonperforming Loans, Banks with Total Assets from \$1B to \$10B, Pacific Census Division (DISCONTINUED) and has a frequency of Quarterly, End of Period. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1988-01-01 and the observation end date is 2020-07-01. The popularity of this series is 1.

5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_LLRNPT39	R-squared:	0.158
Model:	OLS	Adj. R-squared:	0.119
Method:	Least Squares	F-statistic:	4.120
Date:	Thu, 24 Sep 2026	Prob (F-statistic):	0.0546
Time:	15:59:56	Log-Likelihood:	-110.56
No. Observations:	24	AIC:	225.1
Df Residuals:	22	BIC:	227.5
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	152.7995	40.101	3.810	0.001	69.635	235.964
value_fred_SWSTFRBCIRGSP	-0.0016	0.001	-2.030	0.055	-0.003	3.37e-05

Omnibus:	4.915	Durbin-Watson:	0.325
Prob(Omnibus):	0.086	Jarque-Bera (JB):	3.660
Skew:	-0.955	Prob(JB):	0.160
Kurtosis:	3.111	Cond. No.	4.07e+05

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

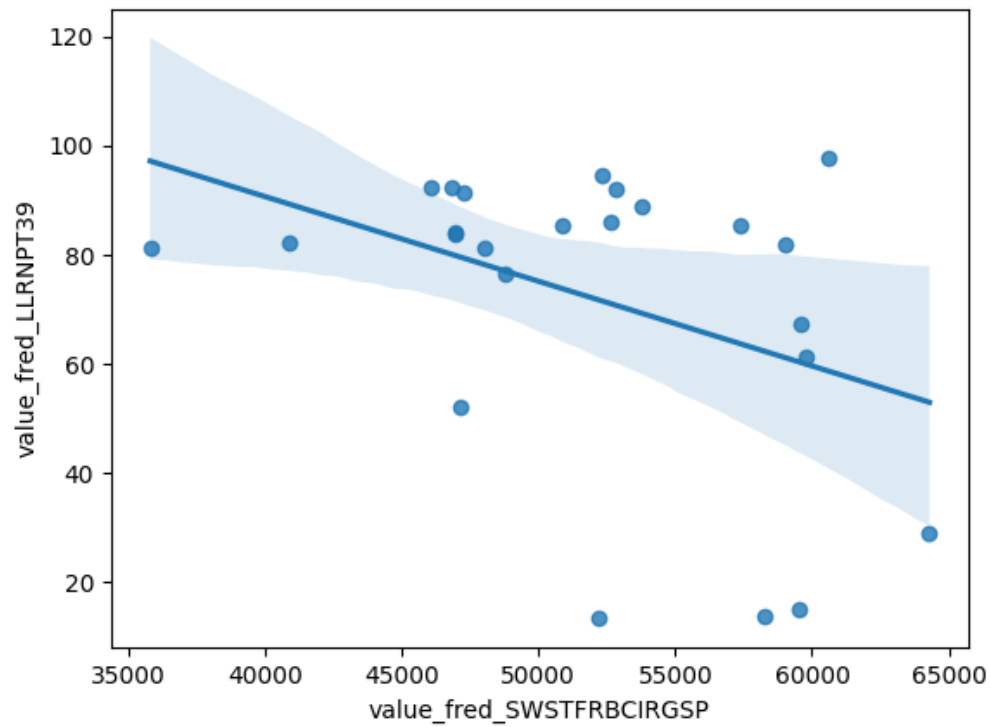


Figure 7: Raw Regression Plot for 2026-09-24

[2] The condition number is large, $4.07e+05$. This might indicate that there are strong multicollinearity or other numerical problems.

5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_LLRNPT39_detrended	R-squared:	0.277
Model:	OLS	Adj. R-squared:	0.244
Method:	Least Squares	F-statistic:	8.412
Date:	Thu, 24 Sep 2026	Prob (F-statistic):	0.00830
Time:	15:59:56	Log-Likelihood:	-108.67
No. Observations:	24	AIC:	221.3
Df Residuals:	22	BIC:	223.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.199e-12	4.776	2.51e-13	1.000	-9.904	9.904
value_fred_SWSTFRBCIRGSP_detrended	-0.0031	0.001	-2.900	0.008	-0.005	-0.001

Omnibus:	7.727	Durbin-Watson:	0.515
Prob(Omnibus):	0.021	Jarque-Bera (JB):	5.892
Skew:	-1.175	Prob(JB):	0.0526
Kurtosis:	3.608	Cond. No.	4.43e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 4.43e+03. This might indicate that there are strong multicollinearity or other numerical problems.

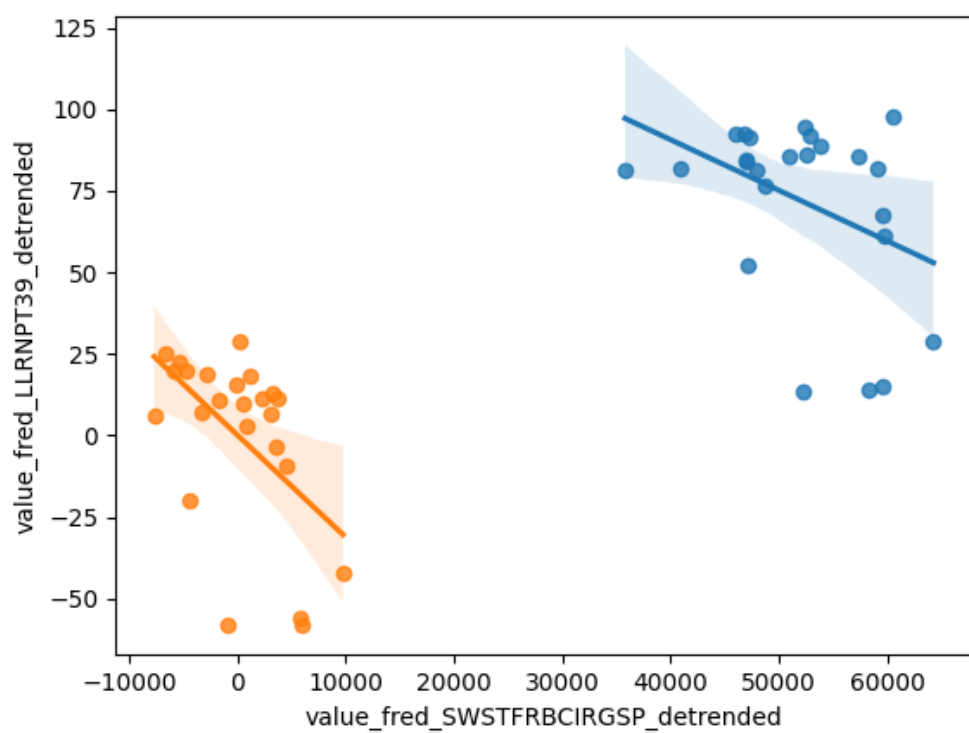


Figure 8: Detrended Regression Plot for 2026-09-24

6 Date: 2026-09-25

Series ID: CWSR0000SAE

This series is titled Consumer Price Index for All Urban Wage Earners and Clerical Workers: Education and Communication in U.S. City Average and has a frequency of Monthly. The units are Index Dec 1997=100 and the seasonal adjustment is Seasonally Adjusted. The observation start date is 1993-01-01 and the observation end date is 2026-08-01. The popularity of this series is 1.

Series ID: LT65EXMNM35A647NCEN

This series is titled Total Tax Exemptions Under Age 65 for New Mexico and has a frequency of Annual. The units are Number of Exemptions and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1989-01-01 and the observation end date is 2023-01-01. The popularity of this series is 1.

6.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_LT65EXMNM35A647NCEN	R-squared:	0.319
Model:	OLS	Adj. R-squared:	0.296
Method:	Least Squares	F-statistic:	13.61
Date:	Fri, 25 Sep 2026	Prob (F-statistic):	0.000923
Time:	16:02:20	Log-Likelihood:	-383.39
No. Observations:	31	AIC:	770.8
Df Residuals:	29	BIC:	773.7
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.111e+06	8.56e+04	12.985	0.000	9.36e+05	1.29e+06
value_fred_CWSR0000SAE	2729.6493	739.846	3.689	0.001	1216.494	4242.805

Omnibus:	1.749	Durbin-Watson:	0.439
Prob(Omnibus):	0.417	Jarque-Bera (JB):	1.177
Skew:	0.477	Prob(JB):	0.555
Kurtosis:	2.985	Cond. No.	937.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

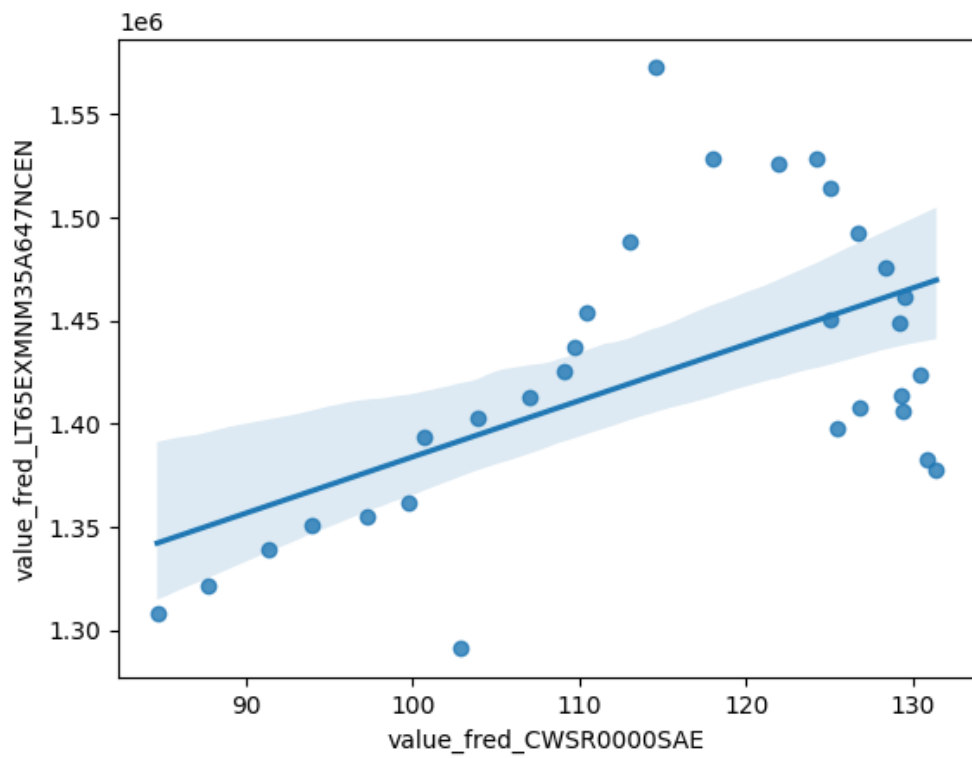


Figure 9: Raw Regression Plot for 2026-09-25

6.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_LT65EXMNM35A647NCEN_detrended	R-squared:	0.508
Model:	OLS	Adj. R-squared:	0.491
Method:	Least Squares	F-statistic:	29.99
Date:	Fri, 25 Sep 2026	Prob (F-statistic):	6.75e-06
Time:	16:02:20	Log-Likelihood:	-375.82
No. Observations:	31	AIC:	755.6
Df Residuals:	29	BIC:	758.5
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	4.538e-09	8271.954	5.49e-13	1.000	-1.69e+04	1.69e+04
value_fred_CWSR0000SAE_detrended	1.073e+04	1959.312	5.477	0.000	6722.977	1.47e+04

Omnibus:	6.655	Durbin-Watson:	0.863
Prob(Omnibus):	0.036	Jarque-Bera (JB):	6.187
Skew:	0.543	Prob(JB):	0.0453
Kurtosis:	4.900	Cond. No.	4.22

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

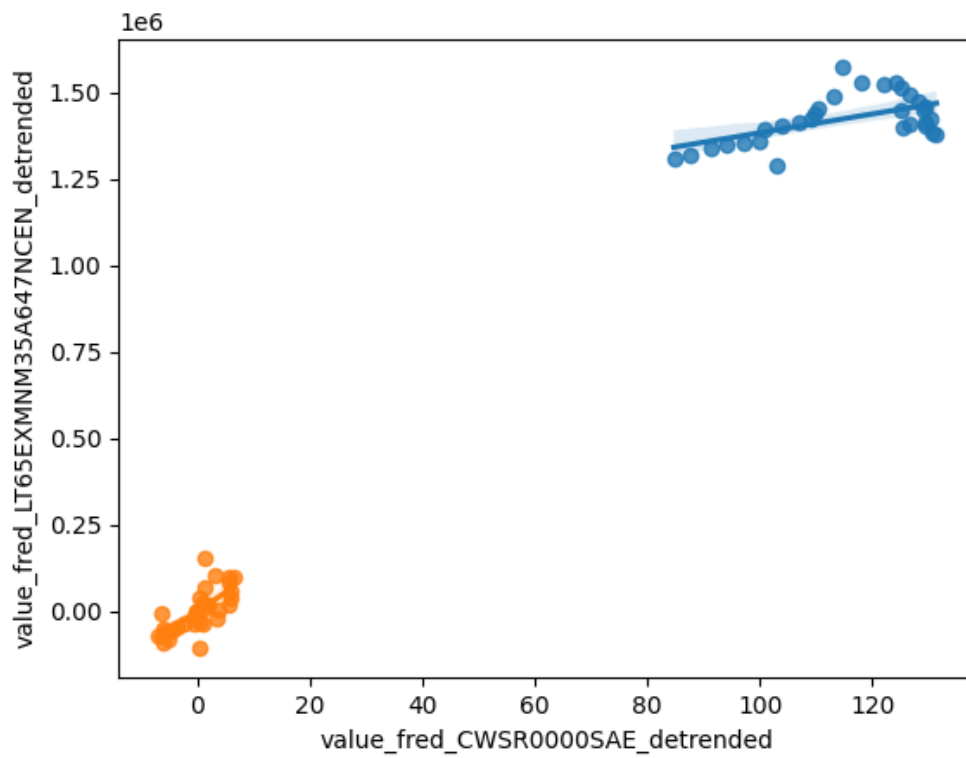


Figure 10: Detrended Regression Plot for 2026-09-25